When Addition Starts Floating: Theta-Links, Log-Links, and Arithmetic Deformation

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§1.1 A multiplicative period and its arithmetic scale

Imagine a curve whose local points can be described by multiplying nonzero numbers, with one special number q acting as a period. Over a nonarchimedean local field L, an elliptic curve with split multiplicative reduction admits a Tate uniformization

$$E_q(\overline{L}) \simeq \overline{L}^{\times}/q^{\mathbb{Z}}, \qquad 0 < |q|_v < 1.$$

The quotient identifies u, qu, q^2u and so on, just as $\mathbb{R}/\mathbb{Z}$ identifies numbers differing by an integer. The operation here is multiplication, though, and the valuation $v(q)$ measures the size of the period in the nonarchimedean direction. It also records the valuation of the minimal discriminant of the curve.

A genuine example gives the notation some weight. Dupuy and Hilado construct initial theta data beginning with the elliptic curve $11a1$,

$$E: \quad y^2 + y = x^3 - x^2 - 10x - 20, \qquad v_{11}(q_{11}) = 5.$$

The valuations of q_{11}, q_{11}^4 and q_{11}^9 are consequently

$$\boxed{5, \quad 20, \quad 45.}$$

Here the squares $1, 4, 9$ are particularly suggestive. Special values of nonarchimedean theta-functions naturally produce q^{m^2}-type information; a theta-link is designed to compare this selected multiplicative information across arithmetic settings.

There is a catch with far-reaching consequences. A comparison that respects products and special theta-values has no automatic reason to respect sums. Once the additive structures can no longer be identified directly, it becomes necessary to control what survives: roots and Galois actions, logarithms of units, and the ambiguities in reconstructing numbers from those records. This is the motivation for theta-links, Kummer theory, log-links and the network of Hodge theaters. Their common goal is a numerical inequality extracted from arithmetic information that can still be compared.[1]

§1.2 The nonarchimedean theta-function

Let L be a local field of characteristic zero with finite residue field, and let C_L be the completion of an algebraic closure of L. Let $q \in L$ be a nonzero element of the maximal ideal of $\mathcal{O}_L$. In the application to elliptic curves, q is the Tate parameter q_v.

[1] See Ivan Fesenko, *Arithmetic Deformation Theory via Arithmetic Fundamental Groups and Nonarchimedean Theta-Functions*, and Taylor Dupuy and Anton Hilado, *The Statement of Mochizuki's Corollary 3.12: Initial Theta Data*.

Definition 1.2.1 (Theta-function of type au^m)**.** A holomorphic function $f : C_L^\times \to C_L$, defined over L, is a theta-function of type au^m, where $a \in C_L^\times$ and $m \in \mathbb{Z}$, if

$$f(u) = au^m f(qu).$$

A meromorphic function invariant under $u \mapsto qu$ descends to a function on the Tate quotient

$$C_L^\times / \langle q \rangle.$$

The basic nonarchimedean theta-function used in the survey is

$$\theta(u) = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(n-1)/2} u^n.$$

It also has the product expansion

$$\theta(u) = (1-u) \prod_{n \geq 1} (1-q^n)(1-q^n u)(1-q^n u^{-1}),$$

which is the nonarchimedean analogue of the Jacobi triple product expression. The function is chosen because its zeros and poles interact well with the Tate curve and with the theta-values used in IUT.

Remark 1.2.2 (Why a theta-function appears) — The theta-function is not added to the story for analogy with complex analysis alone. On a Tate curve, elliptic functions with period q can be expressed in terms of theta-functions. Thus theta-functions are a natural language for functions on the local analytic uniformization

$$E(L) \cong L^\times / \langle q \rangle.$$

Fesenko also records a modified theta-function, written here schematically as $\ddot{\Theta}(u)$, whose special values at points separated by periods from $\pm i$ satisfy a relation of the form

$$q^{m^2/2} \ddot{\Theta}(i\sqrt{q^m}) = \ddot{\Theta}(i).$$

After choosing a $2l$-th root of q, this leads to theta-values of the shape

$$\Theta(\sqrt{-q^m}) = q^{m^2}, \qquad 1 \leq m \leq \frac{l-1}{2},$$

up to multiplication by roots of unity of order dividing $2l$.

Remark 1.2.3 (A controlled simplification) — The formula above is the correct structural feature needed here: special theta-values encode powers q^{m^2}. The complete IUT construction keeps track of coverings, labels, roots of unity, Galois actions, and compatible local-global data. Those details are not being suppressed because they are unimportant; they are being postponed because the first conceptual step is to see why theta-values can serve as deformation data.

§1.3 The local theta-link as a non-ring-theoretic map

At a bad reduction valuation of odd residue characteristic, the main version of the theta-link described by Fesenko revolves around a local monoid-theoretic transformation. In simplified notation it sends

$$q \longmapsto \{\Theta(\sqrt{-q^m}) = q^{m^2}\}_{1 \leq m \leq (l-1)/2},$$

and it sends units to units by the identity map, up to the appropriate torsion ambiguity.

Definition 1.3.1 (Local theta-link, structural form)**.** A local theta-link is a monoid-theoretic comparison which, at a bad split multiplicative valuation, keeps the unit part essentially fixed while replacing the Tate-period generator q by a family of theta-values behaving like q^{m^2}. It is a comparison of multiplicative and theta-theoretic data, not a homomorphism of local rings.

Thus the local theta-link acts on two kinds of local data:

$$\mathcal{O}_L^\times / \operatorname{Tor}(\mathcal{O}_L^\times), \qquad \{q^{m^2} : 1 \le m \le (l-1)/2\},$$

and it is glued to global monoid-theoretic data satisfying the product formula.

We can feel the difference between a multiplicative comparison and a ring map without knowing IUT. Work just in $\mathbb{Q}_5^\times$. Every nonzero element has a unique expression

$$x = 5^n u, \qquad n \in \mathbb{Z}, \quad u \in \mathbb{Z}_5^\times.$$

Define the following *toy* transformation, with exponent $4 = 2^2$:

$$T(5^n u) = 5^{4n} u.$$

It behaves impeccably with respect to multiplication:

$$T(5^n u \cdot 5^m v) = 5^{4(n+m)} uv = T(5^n u) T(5^m v).$$

It even fixes every unit, so $T(1) = 1$. But consider the entirely ordinary addition $1+5 = 6$. Since 6 is a 5-adic unit,

$$\boxed{T(1+5) = T(6) = 6 \qquad \text{while} \qquad T(1) + T(5) = 1 + 5^4 = 626.}$$

The map remembers products and valuations while failing to preserve sums. Every ring identity still holds *inside* $\mathbb{Q}_5$; it is the attempt to *transport* addition through T that fails.

This calculation is not a construction of a theta-link. It isolates the elementary feature behind Fesenko's explanation: a theta-link compares selected multiplicative/theta-theoretic structures across theaters and provides no ordinary ring isomorphism carrying every sum along with them. The real theory has considerably more data than this toy, but that basic distinction is already unavoidable.

This is the first honest meaning of the phrase "addition starts floating". It does not mean that the equation $1 + 1 = 2$ becomes false inside a fixed field. It means that after theta-deformation one is no longer allowed to identify the additive coordinates on the two sides as if a ring homomorphism had carried them across.

§1.4 The bridge supplied by Kummer theory

If the theta-link only destroyed additive compatibility, it would not be useful. The next ingredient explains how monoid-theoretic data can still communicate with Galois and fundamental-group data.

For an elementary glimpse, suppose a field L contains the ℓ-th roots of unity and we adjoin α with

$$\alpha^\ell = a \in L^\times.$$

Any Galois automorphism σ fixing L must send α to another root of the same equation. Therefore

$$\sigma(\alpha) = \zeta_\sigma \alpha, \qquad \zeta_\sigma^\ell = 1, \quad \text{or} \quad \boxed{\frac{\sigma(\alpha)}{\alpha} \in \mu_\ell.}$$

One piece of *multiplicative* information, the possibility of extracting an ℓ-th root, has produced a record of *Galois* action. For example, when $\alpha = q^{1/\ell}$, the possible ratios $\sigma(\alpha)/\alpha$ explain why extracting roots of a Tate parameter naturally creates torsion labels. If the roots of unity are fixed by the Galois group, the ratios even form a character; in the general situation they form a cocycle.

This is the small calculation behind the much more elaborate generalized Kummer correspondence. The point is to keep enough information about roots and Galois symmetries to compare structures even when an ordinary additive identification is unavailable.

Definition 1.4.1 (Kummer map, schematic form)**.** Let H be an open subgroup of a Galois or arithmetic fundamental group acting on an abelian group M. Kummer theory studies maps of the form

$$M^H \longrightarrow H^1(H, \mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, M)),$$

obtained by considering divisibility of elements of M.

For ordinary intuition, Kummer theory turns the question of extracting roots into a cohomological Galois datum. In IUT, the relevant versions are generalized and truncated. They are applied not merely to elements of fields, but also to monoid-theoretic structures associated with theta-functions, theta-values, number fields, and their completions.

Remark 1.4.2 (Kummer theory as a translator) — The theta-link is monoid-theoretic. Mono-anabelian reconstruction works on the side of arithmetic fundamental groups. Kummer theory is one of the translators between these languages. It relates theta-values and monoid data to Galois or fundamental-group structures that can be reconstructed and compared.

A key issue here is cyclotomic rigidity. Various copies of $\widehat{\mathbb{Z}}$, or quotients of such copies, arise from roots of unity, from monoid-theoretic data, and from subquotients of fundamental groups. IUT needs algorithms that identify the relevant cyclotomic structures in a canonical way. Without such rigidity, the roots of unity introduced by extracting l-th or $2l$-th roots would create uncontrolled ambiguity.

§1.5 Two symmetries and the bookkeeping of labels

The theta-link is built after choosing a prime $l > 3$. The l-torsion points of the elliptic curve provide labels modeled on $\mathbb{F}_l$. Fesenko emphasizes two different symmetries.

Definition 1.5.1 (The two label symmetries)**.** The geometric-additive symmetry is

$$\mathbb{F}_l^{\times\pm} = \mathbb{F}_l \rtimes \{\pm 1\},$$

acting on labels by

$$z \longmapsto \pm z + a.$$

The arithmetic-multiplicative symmetry is

$$\mathbb{F}_l^{*\pm} = \mathbb{F}_l^{\times}/\{\pm 1\}.$$

The first symmetry is essentially geometric. It is related to the geometric part of the arithmetic fundamental groups and to Kummer theory around theta-values. It is also

used to synchronize conjugacy ambiguities of local Galois groups attached to different labels.

The second symmetry is essentially arithmetic. It is related to the Galois action of number fields and is multiplicative by definition. It helps separate the zero label from the nonzero labels and is more naturally multiradial.

Symmetry	Nature	Role
$\mathbb{F}_l^{\times\pm}$	geometric/additive	conjugate synchronization
$\mathbb{F}_l^{*\pm}$	arithmetic/multiplicative	label bookkeeping and descent

Remark 1.5.2 (Why labels are not cosmetic) — The labels are not merely indices in a long construction. They keep track of different copies of local Galois data and theta-values. If these labels are identified too naively, one loses the distinction between the geometric-additive and arithmetic-multiplicative parts of the construction.

§1.6 Conjugate synchronization and the basepoint problem

Fundamental groups depend on basepoints. In ordinary expositions this dependence is often hidden because different basepoints lead to groups identified up to inner automorphism. In IUT this ambiguity cannot simply be ignored, because the theta-link and log-link compare structures that live in different theaters.

Definition 1.6.1 (Conjugate synchronization, informal)**.** Conjugate synchronization is a system of identifications between local absolute Galois groups attached to different labels, arranged so that one does not introduce uncontrolled conjugacy indeterminacy.

The need for such synchronization is one reason the theory becomes much more rigid than a naive "copy-and-paste" of local fields. The comparison is not allowed to choose arbitrary basepoint identifications. It must respect the label symmetries and the Kummer-theoretic data surrounding the theta-values.

§1.7 The Log-Link and Additive Structure

The theta-link uses multiplicative theta-values. To regain access to additive structure, Fesenko turns to the nonarchimedean logarithm.

Definition 1.7.1 (Nonarchimedean logarithm)**.** For a local field L, the nonarchimedean logarithm

$$\log : \mathcal{O}_L^{\times} \longrightarrow L$$

is defined on elements $1-\alpha$, with α in the maximal ideal of $\mathcal{O}_L$, by

$$\log(1-\alpha) = -\sum_{n\geq 1} \frac{\alpha^n}{n},$$

and sends multiplicative representatives of the finite residue field to 0.

The key point is that log transforms multiplication into addition where it is defined. Thus the logarithm lets multiplicative unit data speak an additive language.

One computation is enough to show why a logarithm is the natural partner of multiplication. In $\mathbb{Q}_5$, take units close to 1, say $u, v \in 1 + 5\mathbb{Z}_5$. The convergent power series gives

$$\log_5(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \cdots, \qquad t \in 5\mathbb{Z}_5,$$

and consequently

$$\boxed{\log_5(uv) = \log_5 u + \log_5 v.}$$

For instance, $6 = 1 + 5$ and $36 = 6^2$ belong to this group, so

$$\log_5(36) = 2\log_5(6).$$

No numerical approximation is needed: this is an exact identity of 5-adic numbers.

A useful mental picture is that the logarithm changes the *language* of the data. What used to be multiplication of nearby units becomes addition of their logarithms. It does not magically transport all additions across a theta-link. In particular, the toy transformation T above still sends $1 + 5$ and $T(1) + T(5)$ to different numbers. The log-link helps compare a selected, logarithmically encoded piece of the additive story, while the theory has to control the pieces it cannot identify directly.

Definition 1.7.2 (Log-link, structural form)**.** A log-link is the comparison which shifts multiplicative unit data through the nonarchimedean logarithm, so that additive structures can be reconstructed from logarithmic images of multiplicative structures.

Fesenko explains the relation between the two links as follows. The multiplicative structures on the two sides of the theta-link are related through value-group data. The additive structures are related through unit-group data after applying the log-link.

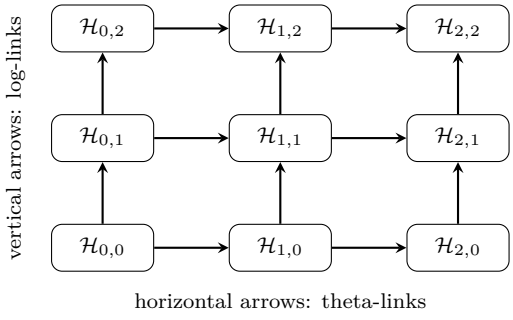
Remark 1.7.3 (Why the logarithm is not optional) — Theta-values naturally act on multiplicative data, not directly on the additive group of the local field. The logarithm creates a setting where multiplicative unit data has an additive image. This is why the log-link is not an auxiliary simplification; it is part of the reconstruction mechanism.

§1.8 The log-theta-lattice

The theta-link and log-link do not appear as isolated arrows. They are organized into a two-dimensional pattern called the log-theta-lattice.

Definition 1.8.1 (Log-theta-lattice, schematic)**.** The log-theta-lattice is a two-dimensional array of theaters. Vertical arrows are log-links, and horizontal arrows are theta-links. A horizontal theta-link at one level depends on the log-link structure immediately below it.

The picture is:



This diagram is only a guide. Fesenko notes that the main results use just two neighboring infinite vertical lines and the horizontal arrow between the lattice points $(0, 0)$ and $(1, 0)$. The important idea is that the theory does not compare a single source and target once. It studies a controlled pattern of comparisons.

§1.9 The Log-Shell

A technical difficulty now appears. The log-link does not commute with the theta-link. If one expected a clean commutative square, one would be disappointed. IUT instead introduces a common structure that can hold the relevant images even when elementwise compatibility fails.

Definition 1.9.1 (Log-shell)**.** For a nonarchimedean local field L, the nonarchimedean part of the log-shell is a compact subgroup of the form

$$(p^*)^{-1} \log(\mathcal{O}_L^\times),$$

where $p^* = p$ if p is odd and $2^* = 4$. At a complex archimedean place, the corresponding log-shell is the closed ball of radius π.

The log-shell should not be imagined as an error bar in a numerical approximation. It is a structural container. The relevant Kummer isomorphisms are not compatible with the log-link at the level of individual elements, but their images are contained in the log-shell. This is exactly the kind of weakened compatibility that arithmetic deformation needs.

Remark 1.9.2 (Compatibility without equality) — In ordinary algebra one often tries to prove that a diagram commutes element by element. The log-shell records a weaker but controlled form of compatibility: one compares inclusions of certain subsets or regions rather than exact equality of individual transported elements.

§1.10 Mono-anabelian reconstruction and multiradiality

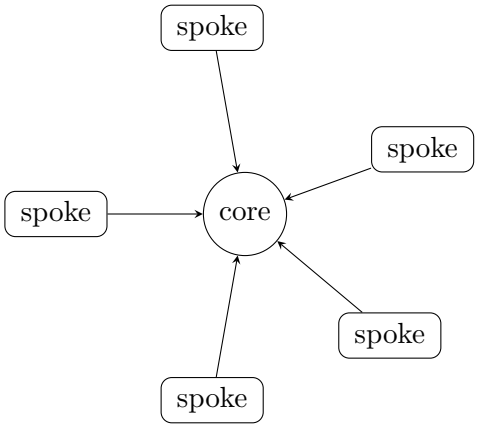
The words “reconstruction” and “multiradiality” are easy to mystify, but their roles are concrete.

Definition 1.10.1 (Mono-anabelian reconstruction)**.** Mono-anabelian reconstruction is the construction of ring-theoretic data from a single topological group that is known to be isomorphic to an arithmetic fundamental group of a suitable hyperbolic curve or orbicurve. The reconstruction is algorithmic and functorial in the sense needed for localization and change of base field.

The difference from ordinary anabelian comparison is important. Classical anabelian geometry often compares two geometric objects through an isomorphism of their fundamental groups. Mono-anabelian geometry asks for algorithms that start from one suitable group and recover the ring structure associated to it.

Definition 1.10.2 (Multiradial algorithm)**.** A reconstruction algorithm is multiradial if an object constructed from one spoke of a wheel-like system is described in terms that still make sense from the viewpoint of other spokes. Equivalently, it is compatible with simultaneous execution at multiple spokes.

Here the "wheel and spokes" language is only an analogy, but a useful one. A spoke corresponds to one theater or one radial perspective. The center corresponds to a core object reconstructed from data that can be described from more than one spoke.



multiradial descriptions must remain meaningful from several spokes at once

Remark 1.10.3 (Why indeterminacy appears) — A multiradial description is stronger than a one-spoke description. To make such descriptions possible, IUT allows certain controlled indeterminacies. These are not defects added after the fact; they are part of how the theory avoids making illegal identifications between different theaters.

§1.11 The three indeterminacies

Fesenko singles out three indeterminacies that enter the computation of volume deformation.

Definition 1.11.1 (The three indeterminacies, structural form)**.** The three main indeterminacies are:

1. Ind1: an indeterminacy related to permutation symmetries of Galois and arithmetic fundamental groups along vertical lines of the log-theta-lattice;
2. Ind2: an indeterminacy related to the horizontal theta-link and to the action of a compact group on $\log(\mathcal{O}_L^\times)$;
3. Ind3: an indeterminacy arising from upper semi-compatibility of Kummer isomorphisms with log-links in a single vertical line.

The third phrase, "upper semi-compatibility", is deliberately weaker than commutativity of a diagram. Instead of asking for equality of transported elements, one asks for controlled inclusion relations among the relevant subsets.

Why keep track of these ambiguities instead of just choosing convenient identifications? Think of trying to compare two measurements when their exact matching is unavailable. One can still compare the *sets of possible results*. For example, if different choices produce

values in $[1, 2]$ and $[3/2, 5/2]$, both sets fit inside the common interval $[1, 5/2]$, even though no particular element of the first interval has been identified with a particular element of the second. The interval is deliberately larger than either set: it is a *container* in which a bound can be stated without pretending to have a pointwise correspondence.

This is only an analogy for the role of a log-shell and the allowed indeterminacies: the actual objects are logarithmic images, Galois-theoretic correspondences and adelic data. Yet it explains the logical advantage of controlling an entire region of possible images. An estimate about its size may survive changes of labels or choices, even when an equality between two chosen elements would make no sense.

The three indeterminacies track different sources of that freedom: the choice of synchronization along the log-links, the symmetries seen across a theta-link, and the weaker compatibility of Kummer-theoretic images with logarithmic data. One should not confuse the *existence of a container* with a proof of the difficult global inequality bounding its size; the latter is the substantive assertion of IUT.

This is where the intuitive phrase "floating addition" becomes mathematically safer. The additive structure is not discarded into chaos. It is reconstructed through logarithmic images and fundamental-group algorithms, while the unavoidable ambiguity is measured by the three indeterminacies.

§1.12 Where the comparison is supposed to lead

We can now follow the story without keeping nine new definitions in our heads. At a split multiplicative place the Tate parameter q records a local scale; theta-values supply the q^{m^2}-type data for the theta-link. The obstacle is already visible in the little map $T(5^n u) = 5^{4n} u$: knowing how to transport products gives no permission to transport sums.

The surrounding mechanisms address different parts of this obstacle. Kummer theory turns selected roots into Galois information. The two symmetries control geometric and arithmetic labels. The log-link expresses multiplication of units in an additive language, while log-shells and indeterminacies hold possible images in a common container. Mono-anabelian reconstruction and multiradial descriptions are meant to keep useful information accessible from several theaters. The log-theta-lattice makes these comparisons interact.

Fesenko's account also gives a strikingly simple piece of algebra behind the *goal* of this machinery. Write $D > 0$ for a normalized degree associated to Tate parameters and V for a logarithmic volume term constructed from the theta-data. The intended inequalities have the shape

$$-D \le V \le a(\ell) - b(\ell) D, \qquad b(\ell) > 1.$$

If these estimates can be established in the relevant arithmetic setting, a school-level rearrangement gives

$$\boxed{(b(\ell) - 1) D \le a(\ell).}$$

The challenge lies in the first comparison: it relates data from different theaters while respecting only the allowed information. Dupuy and Hilado make a useful distinction here. Their computable initial theta data explain how the input to the claimed inequality is assembled; the intricate inter-theater estimate is treated as an assumption, not proved by the input computation.

The phrase *addition starts floating* now has a definite intuition: inside each field addition remains entirely ordinary, but transporting it through a multiplicative comparison

is delicate. The aim is to recover enough bounded, choice-independent numerical information for a Diophantine inequality while keeping the differences between the theaters visible.